

1. AI Integration Workflow

1.1 Why AI is Needed

Traditional blood donation systems use manual matching which is slow and error-prone. In emergencies, finding the right donor quickly can save lives. AI makes this instant, smart, and available 24/7.

Key problems AI solves:

- Instant donor matching based on blood type, location & availability
- Predicting blood shortages before they become critical
- 24/7 intelligent chatbot support for donors and recipients
- Smart notifications to alert nearby compatible donors automatically

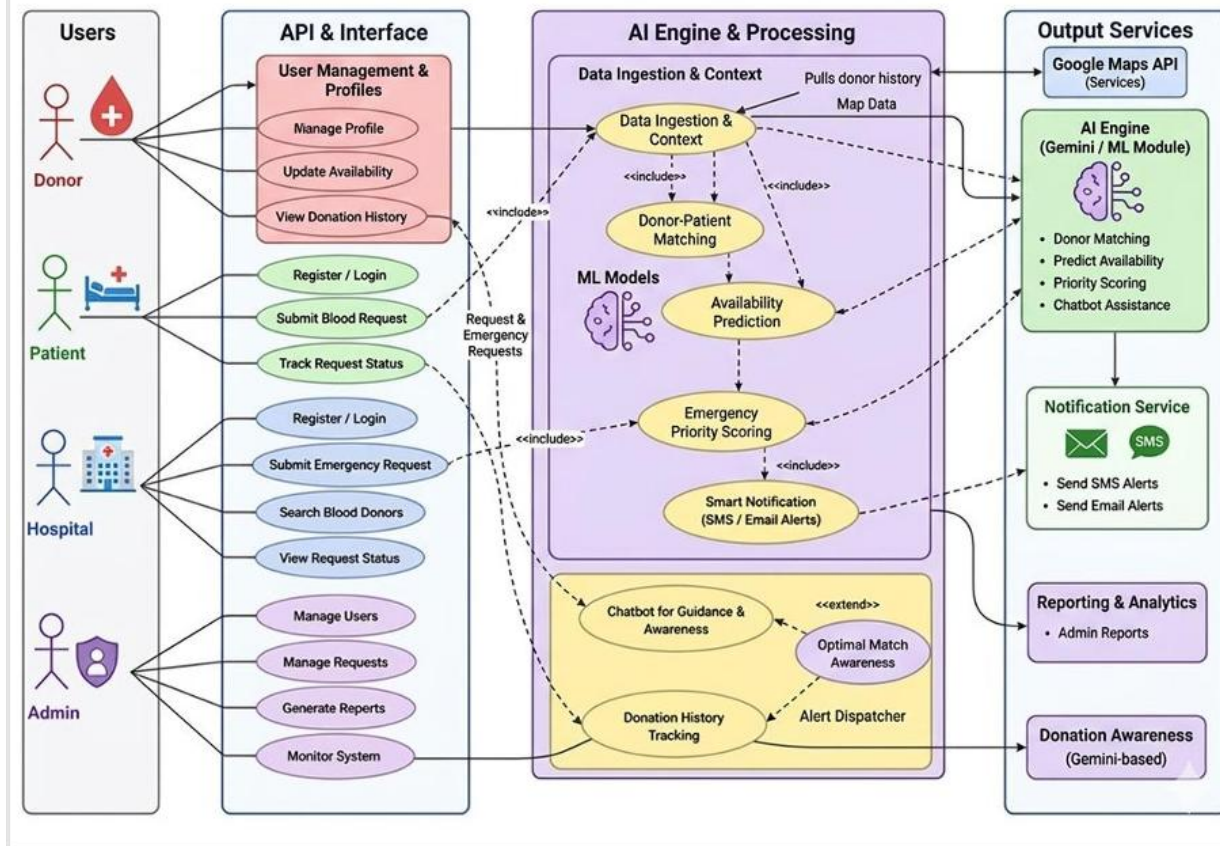
1.2 AI Service / Model Used

Feature	AI Service	Model
Donor Matching	Google Gemini API	gemini-1.5-flash
Chatbot Assistant	Google Gemini API	gemini-1.5-pro
Shortage Prediction	Custom ML Model	scikit-learn / TensorFlow
Compatibility Check	Rule-based AI	Python Logic Engine

1.3 AI Workflow

Complete flow from user request to AI response:

AI Blood Donation Management System: Complete Workflow Diagram



1.4 AI Input & Output

Feature	Input to AI	Output from AI
Donor Matching	Blood group, location (lat/long), urgency, units needed	Ranked donor list with distance, score, contact info
AI Chatbot	User's natural language message, conversation history	Helpful response about eligibility, centers, donation tips
Shortage Prediction	Historical donation data, current stock, date/season	Predicted shortage alert for specific blood groups by region
Compatibility Check	Donor blood group, recipient blood group	Compatible: Yes/No + explanation of compatibility rules

1.5 How AI Improves the System

- Speed: Finds matching donors in seconds instead of manual hours

- Accuracy: Considers blood type, location, availability & health status simultaneously
- Availability: 24/7 chatbot without human staff
- Proactive: Predicts shortages before they occur so hospitals can prepare in advance
- Scalability: Handles thousands of simultaneous requests efficiently

2. AI API Design Document

All AI-related API endpoints for the Blood Donation Management System are listed below with their purpose, input, and expected output.

2.1 POST /api/ai/recommend-donor

Field	Details
Purpose	Find and rank compatible blood donors near the patient's location
Input	{ "blood_group": "B+", "latitude": 22.83, "longitude": 89.55, "urgency": "high", "units_needed": 2 }
Output	{ "donors": [{ "id": 1, "name": "Rahim", "distance_km": 1.2, "score": 95, "phone": "01700..." }], "total_found": 8 }

2.2 POST /api/ai/chat

Field	Details
Purpose	AI chatbot that answers user questions about blood donation, eligibility, and nearby centers
Input	{ "message": "Am I eligible to donate blood?", "conversation_history": [] }
Output	{ "reply": "You are eligible if you are 18-60 years old, weight above 50kg...", "suggested_actions": ["Find donor", "Check eligibility"] }

2.3 GET /api/ai/predict-shortage

Field	Details
Purpose	Predict which blood groups may face shortage in the next 7-30 days in a specific region
Input	Query params: ?region=Khulna&days=7
Output	{ "predictions": [{ "blood_group": "O-", "risk": "high", "current_stock": 5, "predicted_demand": 18 }] }

2.4 POST /api/ai/check-compatibility

Field	Details
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Purpose	Check if a donor's blood group is compatible with the recipient and explain why
Input	{ "donor_blood_group": "O+", "recipient_blood_group": "A+" }
Output	{ "compatible": true, "explanation": "O+ donors can give to A+, B+, AB+ and O+" }

2.5 POST /api/ai/notify-donors

Field	Details
Purpose	AI selects the most suitable nearby donors and sends them automated push notifications
Input	{ "request_id": "REQ001", "blood_group": "A-", "location": { "lat": 22.83, "lng": 89.55 } }
Output	{ "notified_donors": 5, "status": "notifications_sent", "estimated_response_time": "10 mins" }

2.6 API Summary

Method	Endpoint	Purpose	AI Service
POST	/api/ai/recommend-donor	Find best matching donors	Gemini API
POST	/api/ai/chat	Chatbot for user queries	Gemini API
GET	/api/ai/predict-shortage	Blood shortage prediction	ML Model
POST	/api/ai/check-compatibility	Blood compatibility check	Rule Engine
POST	/api/ai/notify-donors	Auto-notify nearby donors	Gemini + Push API